



University of Asia Pacific

Department of Computer Science and
Engineering **CSE 437: Robotics Lab**

LAB REPORT

Experiment Number: 06

Experiment Title: Designing an Internet of Things-based floor cleaning robot

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Date of Submission : 25/04/2026

1. Experiment Name

Designing an Internet of Things-based floor cleaning robot

2. Objective

The objective of this experiment is to design and implement an Internet of Things (IoT)-based floor cleaning robot capable of performing automated cleaning operations while being monitored or controlled remotely. The system integrates sensors, actuators, Arduino microcontrollers, and internet connectivity to enable intelligent navigation, obstacle detection, and cleaning functionality. This project demonstrates the application of IoT in smart home automation and robotics.

3. Apparatus / Hardware & Software Requirements

Hardware Components

- Arduino UNO (2 Units)
- L293D Motor Driver IC
- DC Gear Motors (3–4)
- Ultrasonic Sensor (HC-SR04)
- Servo Motor
- Cleaning Brush / Roller Motor
- Buzzer
- LED Indicators
- Battery Pack
- Wheels & Robot Chassis
- Jumper Wires
- Breadboard

Software Requirements

- Arduino IDE
- IoT Platform (Blynk / IoT App / WiFi Module Control)
- Embedded C / Arduino Programming

4. Circuit Diagram / Schematic

System consists of:

- One Arduino controlling movement and sensors.
- Another Arduino handling cleaning mechanism / communication.
- Ultrasonic sensor detects obstacles.
- Motor driver controls wheel motors.
- Servo motor adjusts cleaning direction.
- Cleaning motor rotates brush for floor cleaning.

Pin Configuration

Ultrasonic Sensor → Arduino

Sensor Pin Arduino Pin

VCC 5V

GND GND

Trig D9

Echo D10

Servo Motor

Servo Wire	Arduino
VCC	5V
GND	GND
Signal	D6

L293D Motor Driver

L293D Pin Connection

Enable 1 D5

IN1 D2

IN2 D3

Enable 2 D11

IN3 D4

IN4 D7

Motor Supply Battery +

Logic Supply 5V

GND Pins Common Ground

Cleaning Brush Motor

- Connected through motor driver output channel.

Indicators

- LED → Status indication
- Buzzer → Obstacle alert

Working Principle

The floor cleaning robot operates using IoT-based automation. The ultrasonic sensor continuously measures the distance between the robot and surrounding obstacles. When an obstacle is detected, the Arduino processes the sensor data and changes the robot's direction automatically.

DC motors drive the wheels through the L293D motor driver, allowing forward, backward, and turning movement. A servo motor controls the cleaning head or direction of cleaning, while a rotating brush motor performs the cleaning operation.

Through IoT connectivity, the robot can be monitored or controlled remotely using a smartphone or web interface, enabling smart cleaning functionality.

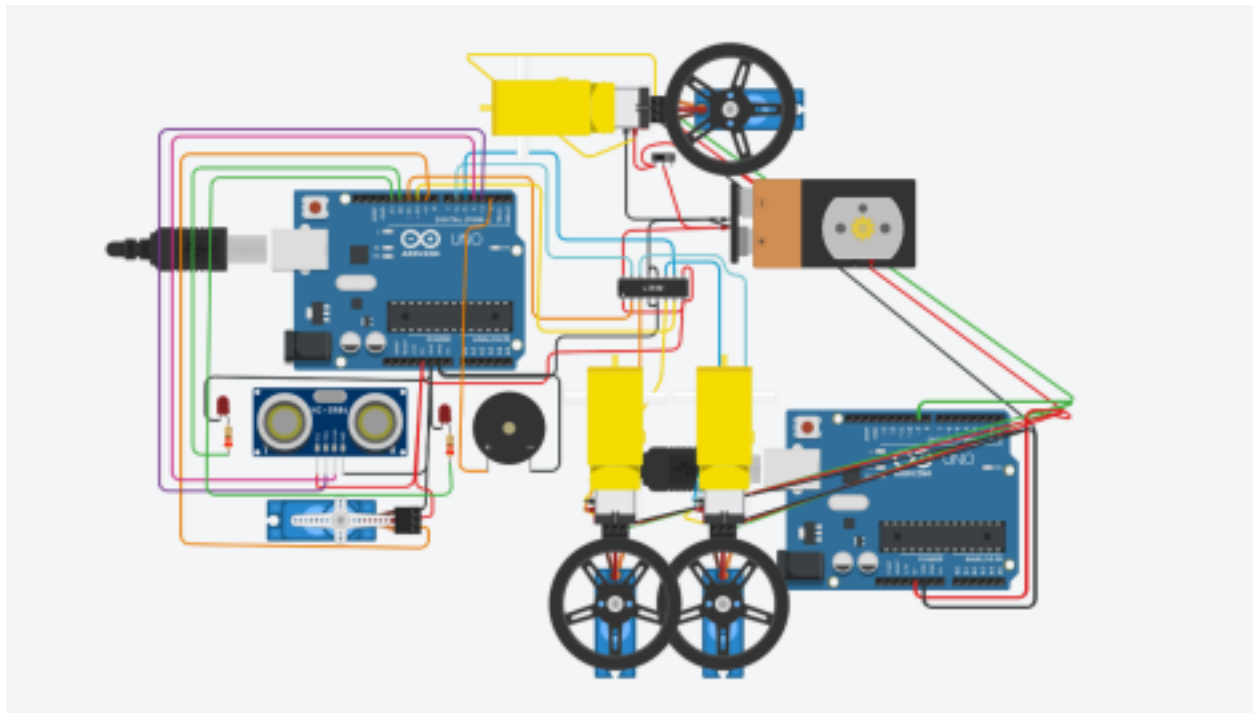


Fig 4.1: Circuit Diagram for Floor Cleaning Robot.

5. Code / Assembly Program

```
#include <Servo.h>
```

```
#define trigPin 9
#define echoPin 10

Servo cleaner;

int ENA = 5;
int IN1 = 2;
int IN2 = 3;
int ENB = 11;
int IN3 = 4;
int IN4 = 7;

void setup() {
  Serial.begin(9600);

  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

  pinMode(ENA, OUTPUT);
  pinMode(ENB, OUTPUT);

  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  cleaner.attach(6);
}

long distance() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH);
  return duration * 0.034 / 2;
}

void forward() {
  digitalWrite(IN1,HIGH);
  digitalWrite(IN2,LOW);
  digitalWrite(IN3,HIGH);
```

```
digitalWrite(IN4,LOW);  
}
```

4

```
void stopRobot() {  
  digitalWrite(IN1,LOW);  
  digitalWrite(IN2,LOW);  
  digitalWrite(IN3,LOW);  
  digitalWrite(IN4,LOW);  
}
```

```
void loop() {
```

```
  long d = distance();
```

```
  if(d < 20) {  
    stopRobot();  
    delay(500);  
  }  
  else {  
    forward();  
  }
```

```
  cleaner.write(90); // cleaning position  
}
```

6. Output / Observations

- Robot successfully moved on floor surface.
- Ultrasonic sensor detected obstacles accurately.
- Robot automatically avoided collisions.
- Cleaning brush rotated continuously during movement.
- IoT system allowed monitoring/control through mobile interface.
- LED and buzzer indicated obstacle detection.

7. Result

The IoT-based floor cleaning robot was successfully designed and implemented. The robot demonstrated autonomous navigation, obstacle avoidance, and cleaning capability with remote monitoring support.

8. Conclusion

This experiment shows how Internet of Things technology can be integrated with robotics to develop smart home automation systems. The designed robot reduces manual cleaning effort and demonstrates real-world applications of embedded systems, sensors, and IoT-based intelligent automation.

